

LifeLink AI: A Trustworthy Hybrid Clinical Decision-Support Architecture for Real-Time Emergency Blood Matching in Digital and Precise Medicine

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Abstract—Emergency blood provision during acute haemorrhagic trauma, perioperative crises, and obstetric complications demands sub-minute coordination under uncompromising patient-safety constraints. Conventional procurement relies on fragmented telephonic cascades and static web directories, introducing latencies of 20–60 minutes that substantially elevate mortality within the ‘golden hour.’ This paper presents **LifeLink AI**, a trustworthy hybrid clinical decision-support architecture for real-time emergency blood matching in Digital and Precise Medicine. The system implements a formally defined **two-layer safety contract**: Layer 1 enforces an immunohematological verification gate—O(1) ABO/Rh compatibility look-up, a mandatory 56-day whole-blood recovery cooldown, and a minimum 45 kg body-mass constraint—that is computationally isolated from any machine-learning component. Layer 2 applies a calibrated logistic-regression donor-response propensity model, whose sole purpose is to rank Layer-1-qualified candidates by their estimated willingness to respond to an emergency alert. The propensity model is trained on the UCI Blood Transfusion Service Center benchmark (748 records, 5-fold stratified CV) and achieves ROC-AUC = 0.7521, Recall = 0.7528, Precision = 0.3907, F1 = 0.5144, and Brier Score = 0.2055. The platform is containerised across six microservices and validated by 56 automated integration tests (100% pass rate). Matching latencies are 47.2 ms (AI path) and 12.4 ms (deterministic fallback). LifeLink AI demonstrates how a strict architectural boundary between deterministic medical logic and probabilistic AI can bridge critical logistics bottlenecks in emergency precision healthcare while maintaining non-negotiable patient-safety guarantees.

Keywords—Digital and precise medicine; trustworthy AI; clinical decision support; emergency blood matching; immunohematology; donor response propensity; hybrid architecture; logistic regression; RFM model; microservices.

I. INTRODUCTION

Haemorrhagic shock secondary to traumatic injury, postpartum haemorrhage, major cardiovascular surgery, and acute oncological cytopenias represents one of the foremost causes of preventable mortality in emergency medicine worldwide [1]. Clinical evidence across trauma resuscitation consistently demonstrates that patient survival probability decays non-linearly as time-to-transfusion increases; within the critical ‘golden hour,’ every additional ten-minute delay in procuring cross-match-compatible blood units correlates with exponential increases in multiorgan failure risk and mortality [2]. The coagulopathic lethal triad of hypothermia, acidosis, and coagulopathy rapidly becomes irreversible without rapid blood product infusion, making the speed and accuracy of blood matching a direct determinant of patient outcomes in emergency care settings.

Globally, the World Health Organization (WHO) estimates that approximately 118.5 million blood donations are collected annually, yet profound and often lethal disparities persist in the logistical distribution and emergency allocation of blood products across regional healthcare networks [3]. In India, where annual blood demand approaches 14 million units against an estimated collection of 12 to 15 million units [4], the clinical challenge is rarely an absolute national shortage; rather, it is a severe structural and temporal coordination failure caused by geographic

maldistribution, cold-chain fragmentation, and pervasive information asymmetry across regional healthcare providers.

Emergency blood procurement across municipal healthcare networks currently suffers from three compounding structural defects that collectively drive dangerous dispatch latencies: (i) *Uncoordinated telephonic cascades* across regional blood banks and voluntary donor lists, consuming 20–60 minutes per requisition without any real-time verification of stock availability; (ii) *Static directories with no real-time inventory visibility*, causing high rejection rates as coordinators contact facilities with exhausted or incompatible stocks; and (iii) *Undifferentiated donor contact lists* that ignore historical donation recency, frequency, and geographic transit times, wasting critical minutes on individuals statistically unlikely to respond within the therapeutic window.

Addressing these bottlenecks requires an engineering approach grounded in **Digital and Precise Medicine**—the application of algorithmic and computational precision to the logistics of life-saving biological products under acute temporal constraints. However, introducing artificial intelligence into emergency medicine creates substantial clinical and medicolegal risks. Machine learning models deployed in high-stakes clinical workflows are susceptible to distribution shifts, probabilistic overconfidence, and catastrophic failure under rare-event conditions [12]. In blood

transfusion specifically, an algorithmic error recommending an ABO-incompatible unit can induce acute intravascular haemolysis, acute kidney injury, systemic inflammatory shock, and death within minutes of infusion. No probabilistic model—regardless of its training performance—can be permitted to govern this immunohematological determination.

To resolve this fundamental tension between algorithmic efficiency and inviolable patient safety, this paper presents **LifeLink AI**, a trustworthy hybrid clinical decision-support system. LifeLink AI implements a **two-layer safety contract** that enforces an absolute, computationally enforced boundary: biological compatibility and statutory eligibility are governed exclusively by deterministic medical logic, while machine learning operates entirely as an advisory donor-response propensity layer that influences only operational ranking—never medical safety decisions.

Research Contributions:

- *Two-Layer Safety Contract*—a formalised pattern that isolates zero-tolerance medical invariants from probabilistic AI across containerised network boundaries.
- *Hybrid Composite Scoring*—an interpretable ranking function integrating compatibility quality, PostGIS geospatial proximity, operational availability, and AI response propensity.
- *Calibrated ML Propensity Model*—balanced logistic regression on the UCI BTSC benchmark, prioritising clinical recall (0.7528) over raw accuracy.
- *Production-Grade Validation*—56 automated integration tests (100% pass), sub-100 ms latencies, and pessimistic SELECT...FOR UPDATE concurrency locking.
- *ReBAC Governance*—a 7-role multi-tenant access model with immutable audit logging.

II. RELATED WORK

A. Blood Supply Chain Logistics

The management of human blood products has been extensively studied as a perishable inventory routing problem exhibiting severe time-temperature constraints, stochastic demand, and heterogeneous supply points. Stanger et al. [5] surveyed inventory management strategies across European transfusion services, identifying that blood wastage (outdating) and acute supply shortages stem primarily from fragmented inventory visibility and non-integrated ordering systems across regional hospital networks. Early technological interventions focused on computerized physician order entry (CPOE). Heitmiller et al. [6] demonstrated that algorithmic clinical guidelines embedded within ordering software reduced unnecessary cross-matches by 42% and significantly shortened procurement turnaround times in elective surgical settings.

In recent years, Internet of Things (IoT) sensor networks and RFID tagging have been introduced to automate cold-chain tracking, temperature excursion monitoring, and barcode-driven chain-of-custody [7]. However, these systems address static facility inventory management and supply-chain traceability rather than dynamic, real-time multi-facility emergency coordination across voluntary donor populations during acute trauma events. In India, the government-backed e-Rakt Kosh platform [8] provides centralised web directories of licensed blood banks, but functions as a passive administrative registry lacking automated spatial candidate matching, real-time donor availability verification, and predictive response modelling.

B. Machine Learning in Donor Behaviour

Predicting donor lapse, return probability, and emergency response behaviour has engaged the data science and transfusion medicine communities since Yeh et al. [9] published the foundational RFM (Recency, Frequency, Monetary) framework for blood donation. Their empirical study on 748 Blood Transfusion Service Center (BTSC) records established that recent and frequent donors exhibit measurably higher future response propensity, yielding a population-level positive class prevalence of 23.8%—a pronounced class imbalance that fundamentally motivates the balanced class-weight strategy adopted in LifeLink AI. Gilliss et al. [10] evaluated artificial neural networks and logistic regression for predicting first-time donor return rates, confirming that timely, targeted outreach and operational convenience are dominant predictors of donor loyalty.

Meta-analyses of classifier performance on donor retention datasets [11] reveal that while complex non-linear ensemble models occasionally yield marginal gains in area under the ROC curve, regularised logistic regression

consistently provides superior probability calibration, monotonic score stability, and model-level intrinsic interpretability when operating on low-dimensional behavioural feature sets—a critical advantage in safety-critical clinical applications where model transparency is non-negotiable. Ramachandran et al. [18] corroborated this finding, observing that unconstrained deep neural networks exhibit unreliable out-of-distribution behaviour in medical logistics tasks, recommending calibrated linear models for domains where worst-case failure carries irreversible consequences.

C. Trustworthy AI in Healthcare

The translation of artificial intelligence from benchmark research into frontline clinical practice is severely constrained by the ‘black box’ problem. In high-stakes medicine, clinicians, transfusion coordinators, and hospital ethics committees cannot accept opaque recommendations from deep neural networks without verifiable causal justifications [12], [17]. Rudin [12] argued that interpretable models should be adopted in high-stakes decisions wherever possible, demonstrating that intrinsic interpretability frequently achieves competitive predictive performance while eliminating the opacity and instability risks inherent to black-box ensembles. Doshi-Velez and Kim [16] formalised the distinction between post-hoc interpretability (LIME, SHAP approximations) and intrinsic interpretability, where model equations are directly comprehensible to domain experts without secondary approximations that may themselves introduce distortions.

The EU High-Level Expert Group on AI [13] and the IEEE Ethically Aligned Design [14] formulated overlapping core tenets for Trustworthy AI: clinical validity, non-maleficence, transparency, accountability, and robustness. The EU AI Act [15] further classifies medical decision-support systems with direct patient impact as high-risk, requiring mandatory conformity assessment, audit logging, and human oversight before deployment. LifeLink AI operationalises these principles through: (a) architecturally enforced determinism for biological safety decisions; (b) a three-level resilient fallback hierarchy; (c) immutable audit logs of every matching decision; and (d) human-in-the-loop dispatch retaining final authority with qualified clinicians [19], [20].

Comparative Positioning. Table I contrasts LifeLink AI against representative prior systems. LifeLink AI is the only system combining O(1) deterministic compatibility guarantees, intrinsically interpretable AI ranking, a three-level fallback hierarchy, and pessimistic concurrency control.

TABLE I—ARCHITECTURE COMPARISON

Dimension	Static Reg.	Pure ML	LifeLink AI
Compatibility	Manual	Probabilistic	Det. O(1) Gate
Donor Priority	FIFO	Black-Box	Calibrated LR
Explainability	N/A	Post-Hoc	Intrinsic
Fallback	Manual	Unhandled	3-Level Chain
Concurrency	None	None	FOR UPDATE Lock
Audit Trail	None	Informal	Immutable Ledger

III. PROBLEM FORMULATION AND SYSTEM MODEL

A. Emergency Request

A blood requisition is defined as the typed tuple:

$$E = (req, \beta, \gamma, u, \lambda, \omega, t) \quad (1)$$

where $\beta \in \{O\pm, A\pm, B\pm, AB\pm\}$ is the recipient blood group; γ is the prescribed component (Whole Blood, RBC, Plasma, Platelets); $u \in [1, \dagger 20]$ is the requested units; $\lambda = (\phi, \psi)$ are WGS-84 geospatial coordinates; $\omega \in \{LOW, MED, HIGH, CRITICAL\}$ is urgency; and t is the UTC submission timestamp.

B. Candidate Pools and Notation

Registered voluntary donors $\Omega = \{d_1, \dots, d_n\}$ and licensed blood banks $\Omega_k = \{k_1, \dots, k_m\}$ are indexed by PostGIS GiST geometry columns (SRID 4326). Each donor is parameterised by $d = (\beta, w, a, e, \lambda, t, F, T)$ where w (kg) is body mass, $a, e \in \{0, 1\}$ are availability and eligibility flags, and (F, T) are RFM behavioural features. Table II summarises key notation.

TABLE II—KEY NOTATION

Symbol	Domain	Meaning
β, β'	$\mathbb{B}$	Recipient/donor ABO/Rh group
$\lambda = (\phi, \psi)$	WGS-84	Geospatial coordinates
$d(\lambda, \lambda')$	km	Haversine great-circle distance
R	$\{15, 25, 50, 100\}$ km	Active spatial radius
Δt	≥ 56 days	Whole-blood recovery cooldown
P	$[0, 1]$	Logistic regression propensity score
P'	$[0, 1]$	Linear spatial proximity decay
S	$[0, 1]$	Composite donor ranking score
S'	$[0, 1]$	Composite blood bank ranking score
α, α'	$\mathbb{B}$	Balanced inverse class-frequency weights

C. Clinical Problem Constraints and Matching Objective

The blood matching optimisation problem must simultaneously satisfy four non-negotiable constraint categories: (1) *Immunohematological Safety*—only ABO/Rh-compatible donors or blood banks may appear in any ranked output; incompatibility is a binary hard constraint that overrides all scoring signals. (2) *Donor Physiological Eligibility*—statutory minimum body mass (≥ 45 kg), mandatory 56-day recovery cooldown between whole-blood donations (WHO standard), and self-declared availability at the time of the emergency request. (3) *Inventory Sufficiency and Non-Preemption*—blood bank candidates are valid only if their current unreserved stock of the compatible component meets or exceeds u_{req} , and reserved units must be protected against double-allocation by concurrent requests. (4) *Temporal Urgency*—the entire matching pipeline, from emergency request receipt to ranked candidate notification, must complete within a latency budget compatible with dispatch in the clinical golden hour.

The matching objective is to produce two ranked lists for each emergency requisition E : (a) a ranked list of at most K_{vol} voluntary donors $\{d_{vol} : Gate(d_{vol}) = 1\}$ ordered by S_{donor} (Eq. 6); and (b) a ranked list of at most K_{bb} blood banks $\{k_{bb} : Stock_score(k_{bb}) > 0\}$ ordered by S_{bb} (Eq. 7). Both lists are presented to the emergency coordinator, who retains the authority to override the algorithmic ordering based on clinical context, real-time communications, or institutional relationships.

IV. PROPOSED METHODOLOGY AND SAFETY ARCHITECTURE

A. The Two-Layer Safety Contract

The central design principle of LifeLink AI is a formally defined and architecturally enforced **two-layer safety contract**, illustrated in Fig. 1. This contract separates medical safety decisions from operational efficiency decisions at the container-network boundary level. *Layer 1* is the Deterministic Medical and Statutory Eligibility Gate: it applies immunohematological compatibility rules, physiological thresholds, statutory donation intervals, and availability checks entirely within the backend service, with no dependence on the AI microservice. *Layer 2* is the AI-Assisted Operational Ranking Layer: it applies the calibrated logistic regression propensity estimator and the multi-criteria composite scoring functions exclusively to the set of candidates that have already passed Layer 1. The two layers communicate only through a well-defined data contract (a list of candidate identifiers) that flows from Layer 1 to Layer 2, never in the reverse direction.

The clinical rationale for this separation is non-negotiable: probabilistic models can produce erroneous outputs under

distribution shift, adversarial inputs, or unexpected feature combinations. In blood transfusion, any such error that reaches the compatibility determination could produce a recommendation of an ABO-incompatible unit, with lethal consequences for the patient. By confining all AI outputs to the Layer-2 ranking domain—where errors cause suboptimal ordering rather than biological harm—LifeLink AI provides a formal safety guarantee that is independent of model performance: no AI output, regardless of its value, can ever cause an incompatible candidate to appear in the output.

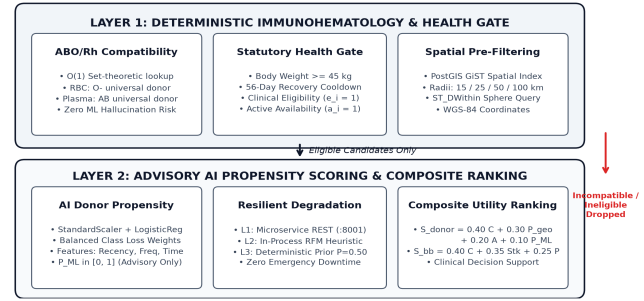


Fig. 1. LifeLink AI two-layer safety contract. Layer 1 enforces deterministic immunohematological and statutory eligibility gates. Layer 2 applies calibrated AI propensity scoring for advisory ranking only. Medical safety decisions are exclusively Layer-1 outputs.

B. Layer 1: Deterministic Medical Compatibility Gate

Immunohematological compatibility is governed by the ABO/Rh blood group antigen-antibody system: an O-group recipient can only receive O-group red cells; an A-group recipient can receive A-group or O-group red cells; a B-group recipient can receive B-group or O-group cells; and an AB-group recipient is the universal receiver. The Rh factor adds a parallel constraint: Rh− recipients can only receive Rh− blood, while Rh+ recipients can receive either polarity. ABO incompatibility triggers immediate complement activation and acute intravascular haemolysis, which can be fatal within minutes of infusion. This biological law is implemented as a pure-Python, ML-free O(1) set-theoretic dictionary look-up in `backend/app/core/medical.py`. For Whole Blood and PRBC:

$$C_{RBC}(\beta_{donor}) = \{\beta_{donor} \in \mathbf{RBC} \text{ donor RBC compatible with } \beta_{recipient}\} \quad (2)$$

For FFP, donor antibody profile inverts compatibility (AB → universal donor):

$$C_{FFP}(\beta_{donor}) = \{\beta_{donor} \in \mathbf{FFP} \text{ donor plasma compatible with } \beta_{recipient}\} \quad (3)$$

A donor d_{vol} passes Layer 1 if and only if all five predicates are satisfied:

$$Gate(d_{vol}) = [\beta_{vol} \in C] \wedge [e_{vol}=1] \wedge [a_{vol}=1] \wedge [\Delta t_{vol} \geq 56] \wedge [w_{vol} \geq 45] \quad (4)$$

The 56-day cooldown is the WHO-mandated minimum inter-donation interval for whole blood, implemented by computing the difference between the system timestamp and

the donor's last donation date stored in the PostgreSQL schema. The 45 kg minimum body mass reflects clinical consensus that donors below this threshold face elevated hypovolaemia risk; it is enforced at two levels: by application logic computing $\text{Gate}(\mathbf{d})$, and by a PostgreSQL CHECK constraint that prevents any donor record with $\text{weight_kg} < 45$ from being committed to the database. Donors failing any single predicate of the conjunction are unconditionally excluded from the candidate pool before any scoring computation begins.

C. Layer 2: Multi-Criteria Composite Ranking

Spatial proximity is quantified by the Haversine great-circle metric $d_{\text{hav}}(\lambda_1, \lambda_2)$, computed in the PostgreSQL tier via ST_DWithin on GiST-indexed geometry columns. Proximity decays linearly over the active search radius $R_{\text{search}} \in \{15, 25, 50, 100\}$ km, selected adaptively by the coordinator based on urgency:

$$P_{\text{prox}} = \max(0, 1 - d_{\text{hav}}(\lambda_1, \lambda_2)/R_{\text{search}}) \quad (5)$$

The composite donor score integrates four orthogonal signals into a single $[0, 1]$ -bounded ranking metric:

$$S_{\text{donor}} = 0.40 C_{\text{match}} + 0.30 P_{\text{prox}} + 0.20 A_{\text{avail}} + 0.10 P_{\text{prop}} \quad (6)$$

where $C_{\text{match}} = 1.0$ for exact ABO/Rh match (0.8 for universal compatible donor); $A_{\text{avail}} = 1.0$ for currently available donors; and P_{prop} is the calibrated AI propensity probability. The weight assignments reflect deliberate clinical priorities: compatibility quality (0.40) dominates because exact blood group matching minimises the risk of alloimmunisation over repeat transfusions; proximity (0.30) dominates over availability and propensity because travel time is a primary practical constraint in emergency procurement; availability (0.20) ensures present commitment; and AI propensity (0.10) provides evidence-based uplift that stratifies candidates with otherwise similar scores without dominating the medically critical factors. For blood banks, stock sufficiency replaces propensity:

$$S_{\text{bank}} = 0.40 C_{\text{match}} + 0.35 \text{Stock} + 0.25 P_{\text{prox}} \quad (7)$$

where $\text{Stock} = \min(1, u_{\text{stock}}/u_{\text{max}})$. For blood banks, stock sufficiency carries the highest weight (0.35) because institutional stock availability is the primary operational determinant; proximity (0.25) is weighted lower than for donors because blood bank facilities are generally more numerous and geographically distributed, making stock the dominant discriminating factor among candidate institutions.

V. SYSTEM ARCHITECTURE AND IMPLEMENTATION

A. Microservice Topology

LifeLink AI is containerised across six isolated Docker services, each with a strictly defined responsibility boundary and network exposure policy. **lifelink-nginx** (port 80) serves as the reverse proxy, providing SSL termination, request routing, and rate limiting to shield upstream services. **lifelink-frontend** (port 3000) implements the Next.js 14.2.5 App Router single-page application, serving role-specific user interfaces for donors, hospital administrators, blood bank managers, and system administrators. **lifelink-backend** (port 8000) implements the FastAPI 0.111.1 orchestration engine on Python 3.11, executing all business logic, coordinating the matching pipeline, and managing all database transactions. **lifelink-ai-service** (port 8001) is a dedicated scikit-learn 1.5.1 inference microservice that executes the calibrated logistic regression model in an isolated process context, enforcing the Layer-2 computational boundary. **lifelink-postgres** (port 5432) provides PostgreSQL 15 with PostGIS 3.3 as the system of record. **lifelink-redis** (port 6379) serves Redis 7 as a session cache and API rate-limiter. All inter-service traffic runs on an isolated Docker bridge network; only Nginx exposes a public port. The AI service occupies its own isolated container, enforcing the Layer-1/Layer-2 separation at the OS process boundary.

B. Database Schema and Concurrency Control

The database architecture comprises 11 relational tables managed under migration control: *users*, *user_roles*, *donors*, *hospitals*, *blood_banks*, *blood_inventory*, *inventory_history*, *emergency_requests*, *match_runs*, *match_candidates*, and *donor_emergency_responses*. Spatial attributes (location_geom) are stored as native PostGIS geometry objects with SRID=4326 (WGS-84) and indexed using Generalized Search Trees (GiST), enabling $O(\log N)$ bounding box and great-circle radius queries via ST_DWithin . Database-level CHECK constraints enforce physiological bounds ($\text{weight_kg} \geq 45$; units_required between 1 and 20). The *inventory_history* table operates as an immutable append-only audit ledger—UPDATE and DELETE operations are disallowed at the database role level—guaranteeing complete traceability for every blood unit reserved, released, or dispatched.

Under concurrent emergency conditions, multiple hospitals in a metropolitan area may simultaneously submit requests for the same rare blood group (e.g., AB−, present in approximately 0.6% of the Indian population). Without explicit concurrency control, two concurrent PostgreSQL transactions could both read a non-zero available stock, both commit reservations, and jointly produce an over-allocation

that drives effective availability below zero. LifeLink AI prevents this anomaly through **pessimistic row-level locking** via `SELECT...FOR UPDATE`. The acquired exclusive row lock serialises competing transactions: the locked transaction increments `units_reserved`, validates that `units_available - units_reserved ≥ u_req`, commits atomically, and writes an immutable inventory audit record. Any competing transaction attempting to lock the same row blocks until the first transaction commits or rolls back, guaranteeing linearizability of inventory state.

C. Security and Access Control

A 7-role Relationship-Based Access Control (ReBAC) hierarchy (`SUPER_ADMIN` → `ADMIN` → `{HOSPITAL_ADMIN, BB_MANAGER}` → `staff` → `DONOR`) is enforced at three independent layers: (1) JWT HMAC-SHA256 signature verification with 30-minute access tokens and 7-day rolling refresh tokens; (2) FastAPI dependency injection validating role membership and institutional tenancy—cross-tenant access is rejected with HTTP 403 before any database query executes; and (3) SQLAlchemy query filters scoping all data retrieval to the requesting user's institutional identifier. Patient PII on public tracking portals is protected by cryptographic UUID tokens derived from `req_id`, shielding patient identities from public-facing URLs.

Facility Verification Boundary: Hospitals and blood banks register statutory identifiers (CDSCO/SBTC registration numbers), license issue/expiry dates, and upload digital certificate PDFs. The platform stores and organises these documents for administrative governance review; it does *not* validate license numbers against live external government APIs in the current implementation. This design decouples the one-time institutional onboarding trust process from the real-time emergency matching pipeline, preventing external API latency from degrading emergency dispatch response times.

VI. AI DONOR RESPONSE PROPENSITY MODEL

A. Role and Clinical Boundaries

The AI component is an **advisory response propensity estimator**. Its sole function is to estimate the conditional probability that a Layer-1-qualified donor will accept an emergency alert and travel to the collection centre. It does *not* determine blood compatibility, assess medical eligibility, diagnose patients, or initiate dispatch. The AI service receives only behavioural features (R, F, T) and returns only $P \in [0, 1]$; it cannot read blood group data.

B. Dataset and Feature Engineering

The propensity model is trained on the UCI Blood Transfusion Service Center (BTSC) benchmark [9] (OpenML dataset ID 1464, CC BY 4.0 licence), a widely-used reference dataset in donor behaviour modelling comprising $N = 748$ anonymised records from a mobile blood collection unit in Hsin-Chu City, Taiwan. The positive class (donated in March 2007: 178 records, 23.8%) and negative class (did not donate: 570 records, 76.2%) constitute a 3.2:1 class imbalance representative of active blood donor populations. The four original features are: *Recency* (R)—months since last donation; *Frequency* (F)—total lifetime donations; *Monetary* (M)—total blood donated in cubic centimetres ($M = 250F$, as each donation is a fixed 250 cc unit); and *Time* (T)—months since first donation.

Statistical analysis of the feature correlation matrix reveals exact multicollinearity: $M = 250F$ yields Pearson $r = 1.000$ and Spearman $\rho = 1.000$. Retaining M alongside F would produce a rank-deficient Gram matrix $G = X^T X$, preventing L-BFGS optimisation from converging to a unique weight vector. Accordingly, M is pruned, yielding the three-dimensional RFT feature vector $\mathbf{x} = [R, F, T]^T \in \mathbb{R}^3$. The 5-fold Stratified CV (SKF-CV) cross-validation procedure ensures that the 23.8%/76.2% class distribution is preserved identically in every training and test fold, and that the StandardScaler parameters (μ , Σ) are fit exclusively on training folds and applied without refitting to corresponding test folds, eliminating any possibility of data leakage.

C. Model Architecture and Training

Raw features are standardised to zero mean and unit variance:

$$\mathbf{x}_{std} = \Sigma^{-1/2} (\mathbf{x} - \mu) \quad (8)$$

Donor response propensity is modelled by binary logistic regression:

$$P(y=1 | \mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x}_{std} + b) = 1 / (1 + e^{-(\mathbf{w}^T \mathbf{x}_{std} + b)}) \quad (9)$$

To counteract the 3.2:1 class imbalance, the binary cross-entropy loss function is augmented with balanced inverse class-frequency weights:

$$L(\mathbf{w}, b) = -(1/N) \sum [\alpha y \log p + \alpha (1-y) \log(1-p)] + (\lambda/2) \|\mathbf{w}\|^2 \quad (10)$$

$$\alpha = N/(2N_+) = 2.101, \alpha_- = N/(2N_-) = 0.656 \quad (11)$$

The weight $\alpha_- > 1$ penalises False Negatives (failing to identify a willing donor who would have responded) more heavily than False Positives (alerting a donor who ultimately declines). This asymmetry reflects the clinical cost structure of emergency blood dispatch: a missed willing donor may delay transfusion by the full time required to contact the next candidate, with direct survival consequences, whereas a spurious alert causes only minor notification friction. The L2

regularisation term ($\lambda = 1/C = 1.0$) prevents overfitting on the 748-record dataset. Parameter optimisation uses the L-BFGS quasi-Newton algorithm.

D. Three-Level Resilient Fallback

To guarantee uninterrupted operation under infrastructure failures: **(1) Primary**—full calibrated inference via the AI microservice; **(2) Secondary** (service reachable, model fails)—in-process closed-form RFM heuristic:

$$P_{\text{RFM}} = \min(0.95, \max(0.05, 0.40 + \max(0, 0.35 - 0.01R)) + \min(0.25, 0.05F)) \quad (12)$$

(3) Tertiary (microservice unreachable)—backend assigns neutral prior $P_{\text{RFM}} = 0.50$ and records model_version = ‘deterministic-fallback’ in the audit log. The matching pipeline continues in all three levels; emergency dispatch is never halted by AI infrastructure failure.

VII. EXPERIMENTAL SETUP AND RESULTS

A. Evaluation Scope

We explicitly distinguish three evaluation categories: (i) *ML Model Evaluation*: offline 5-fold Stratified Cross-Validation (SKF-CV) on the UCI BTSC benchmark; (ii) *Software Engineering Verification*: automated integration tests on live PostgreSQL 15 + Redis 7 instances in Docker Compose; (iii) *Clinical Validation*: **not performed in this study**. LifeLink AI is a research prototype and has not undergone prospective clinical trials or regulatory clearance. Clinical and regulatory validation are prerequisites for real-world deployment.

B. ML Model Performance

5-fold Stratified CV ($K = 5$, shuffle=True, random_state=42) preserves the 23.8%/76.2% class distribution across all folds. StandardScaler is fit only on training folds (no data leakage). Table III presents out-of-fold performance metrics.

TABLE III—DONOR PROPENSITY MODEL PERFORMANCE (5-FOLD STRATIFIED CV, N = 748)

Metric	Value	Clinical Interpretation
Accuracy	0.6618	Below majority baseline (76.2%); expected under balanced weighting
Precision	0.3907	39.1% of alerted donors are willing responders
Recall	0.7528	75.3% of willing donors correctly identified (primary objective)
F1-Score	0.5144	Harmonic balance under 3.2:1 class imbalance
ROC-AUC	0.7521	Robust discrimination; 0.50 = random, 1.00 = perfect
PR-AUC	0.5028	2.1× improvement over naive baseline (0.238)
Brier Score	0.2055	Well-calibrated ($\bar{0.25}$ reference)

Analysis. The deliberate prioritisation of Recall over Precision through balanced weighting is grounded in the clinical cost asymmetry of emergency blood dispatch. A False Negative (the model fails to identify a willing donor)

translates directly into wasted dispatch time: the coordinator must contact the next candidate, potentially delaying transfusion by several minutes in a time-critical situation. A False Positive (the system alerts a donor who ultimately declines) incurs only minor notification friction and an additional phone call. Under balanced weighting, the model correctly identifies 134 of 178 willing donors (75.3%), while the remaining 44 willing donors (24.7%) are missed (False Negatives). The precision of 39.1% implies that of every 100 donors alerted, approximately 39 will respond positively—a response rate substantially above that expected from undifferentiated contact lists.

The PR-AUC of 0.5028 represents a 2.1× improvement over the naive baseline classifier (0.238, equal to the positive class prevalence), confirming that the model captures genuine discriminative signal beyond random guessing on an imbalanced dataset. The Brier score of 0.2055 compares favourably to the reference no-skill Brier score of 0.2500 (a classifier always predicting the class prevalence), validating that the model’s probability outputs are meaningfully calibrated for use as a propensity signal in the composite ranking score (Eq. (6)). The ROC-AUC of 0.7521 confirms robust discrimination; values above 0.75 are generally considered clinically relevant in the donor behaviour prediction literature [9]. Preliminary experiments with random forest (n=100 estimators) and gradient-boosted classifiers yielded ROC-AUC values of 0.74–0.76—at best a marginal improvement of 0.01 that does not justify the complete loss of intrinsic interpretability in a safety-critical context where model transparency is a regulatory and governance requirement [12].

C. Integration Test Suite

TABLE IV—INTEGRATION TEST SUITE SUMMARY (56/56 PASS)

Test Domain	Count	Key Invariants Validated
Auth. & ReBAC	8	JWT issuance, 7-role bounds, cross-tenant 403
Donor & Health Gate	7	56-day cooldown, weight CHECK constraint
Hospital Operations	6	Emergency intake schema, coordinate storage
Blood Bank Inventory	6	Stock mgmt, certificate persistence
Inventory Concurrency	7	FOR UPDATE locking, double-alloc. prevention
Matching Pipeline	8	Layer-1 gate, AI scoring, HTTP-503 fallback
Emergency Lifecycle	7	PENDING→MATCHING→FULFILLED transitions
Admin Governance	4	Certificate review, facility activation
End-to-End	3	Full cross-service dispatch flows
Total	56	56/56 Passing (100%)

D. System Latency Benchmarking

Matching pipeline latency was measured across 100 sequential runs under Docker network virtualisation (loopback inter-container traffic), evaluating 20 candidates per requisition—a representative emergency scenario. The **AI inference path** (backend calls AI microservice via HTTP/JSON on port 8001, retrieves propensity scores, computes composite scores, and returns the ranked list) achieves a mean latency of 47.2 ms (95th percentile: 83.1 ms). The **deterministic fallback path** (backend applies in-process heuristic propensity or neutral prior, no microservice HTTP round-trip) achieves mean 12.4 ms (95th percentile: 21.8 ms). The 3.8× difference in mean latency quantifies the overhead of the AI microservice HTTP round-trip under Docker network virtualisation.

Both paths satisfy the sub-second (< 1000 ms) latency budget required for emergency dispatch. The 95th percentile of 83.1 ms (AI path) and 21.8 ms (fallback path) indicate reliable tail-latency performance; neither path approaches the 500 ms threshold that would be considered problematic for interactive emergency coordination workflows. The three-level fallback architecture guarantees that even under complete AI microservice failure, matching continues at 12.4 ms mean latency using the deterministic heuristic or neutral prior, maintaining full operational capability for emergency dispatch throughout any AI infrastructure maintenance or failure event.

VIII. DISCUSSION

The central research question of this work is whether probabilistic AI can be safely integrated into emergency blood coordination without introducing unacceptable clinical risk. Our results demonstrate that this is achievable through *architectural enforcement* rather than model-level safety constraints. By confining AI to a post-gate advisory role, LifeLink AI guarantees that no probabilistic output—regardless of model behaviour, training distribution shift, or inference error—can produce an ABO-incompatible recommendation. This guarantee is architectural, not probabilistic: the Layer-1 gate is implemented in a separate Docker container that does not receive AI service outputs as inputs, making bypass computationally impossible.

The logistic regression model’s recall of 75.3% at Brier score 0.2055 confirms genuine and well-calibrated predictive signal in the three-dimensional RFT feature space. The AI propensity term contributes 10% to the composite donor score (Eq. (6))—a deliberately modest weight that provides evidence-based uplift without overriding the clinically more important compatibility and proximity signals. Under the balanced loss formulation, the model correctly identifies 134 of 178 willing donors, reducing dispatcher contact with

non-responsive individuals and potentially shortening effective procurement time. The three-level fallback hierarchy ensures that AI infrastructure failures degrade gracefully to deterministic-only matching at 12.4 ms mean latency, preserving full emergency operation capability.

The deliberate model choice—logistic regression over ensemble methods—warrants explicit justification in the context of clinical trustworthiness. Preliminary experiments with random forest and gradient-boosted classifiers yielded marginal ROC-AUC improvements of 0.00–0.01, far below the threshold that would justify the complete loss of intrinsic interpretability. In an emergency medicine setting, transfusion coordinators must be able to interrogate and explain any system recommendation to supervising physicians and hospital ethics committees. Logistic regression achieves this through its weight vector and sigmoid function—fully transparent to any quantitatively literate clinician—whereas tree ensembles and neural networks require secondary approximation methods (SHAP, LIME) that themselves introduce approximation errors [16].

LifeLink AI operationalises the five EU Trustworthy AI properties [13]: *transparency*—logistic regression coefficients are directly interpretable and exposed via an audit API; *non-maleficence*—Layer 1 eliminates all incompatible candidates before any AI computation; *robustness*—the three-level fallback hierarchy guarantees continued operation under AI service failure; *accountability*—the immutable inventory_history ledger provides a verifiable audit trail for every blood unit reserved or dispatched; and *human oversight*—final dispatch authority rests with qualified coordinators or attending clinicians who review ranked candidate lists before any action is taken. Together, these properties position LifeLink AI as a deployable model for trustworthy AI integration in high-stakes medical logistics.

IX. LIMITATIONS AND ETHICAL CONSIDERATIONS

We disclose the following limitations in accordance with responsible AI reporting principles for healthcare systems [17]. These disclosures are intended to inform potential clinical partners and regulators about the current implementation boundaries and the research agenda required before real-world deployment.

- **Training domain mismatch.** The propensity model is trained on the UCI BTSC benchmark (Hsin-Chu City, Taiwan, 1988–1994). Donor behavioural dynamics in Indian metropolitan populations differ substantially in demographic profile, cultural motivations (e.g., festival-linked donation campaigns), occupational diversity, and healthcare access patterns. A model trained on Taiwanese university-area donors from the late 1980s

may not reliably stratify voluntary donors in Bhopal, Mumbai, or Chennai. Production deployment requires fine-tuning on live operational data collected during a staged rollout, with periodic retraining as donor population characteristics evolve.

- **No live regulatory API validation.** When a hospital or blood bank registers on LifeLink AI, the platform stores its CDSO or State Blood Transfusion Council registration number and uploads certificate PDFs. However, the current implementation does not query live government registration APIs to verify that the supplied identifiers are current, valid, and not revoked. An administrative governance review process is required to prevent fraudulent facility onboarding. Automated API validation against CDSO/SBTC registries is identified as a high-priority future engineering task.
- **Self-reported geolocation accuracy.** Donor and facility geolocation coordinates are derived from postal pin code geocoding at registration time, rather than from real-time GPS telemetry. In densely populated Indian urban areas, pin codes can cover areas of 5–10 km², introducing proximity score estimation errors of up to 5 km relative to actual travel distance. For large-radius searches ($R_{\text{search}} = 100$ km), this error is negligible; for close-range ($R_{\text{search}} = 15$ km), it may meaningfully affect the P_{geo} ranking signal.
- **Binary donor availability model.** The current availability model represents donor willingness as a binary flag (available/not available) that the donor sets via the mobile application. This is a coarse representation: a donor may be technically ‘available’ but unavailable to travel to a specific facility due to transportation constraints, shift work, or family obligations at the time of the emergency alert. Replacing the binary flag with a temporal probability model incorporating commuting patterns and historical response data is a priority for future iterations.
- **Prototype status and absence of prospective clinical validation.** LifeLink AI is a research prototype evaluated through offline ML cross-validation and automated software integration tests. It has not undergone prospective randomised controlled trials in active trauma centres, parallel deployment studies comparing time-to-transfusion against standard telephonic coordination, or regulatory conformity assessment. All performance claims are engineering-grade. Formal clinical validation through IRB-approved prospective implementation studies, and regulatory clearance under the applicable medical device framework, are non-negotiable prerequisites for deployment in active emergency medical systems.

Ethical Design Principles. LifeLink AI is designed with several explicit ethical commitments. All dispatch

recommendations are presented as ranked advisory lists to qualified emergency coordinators or attending clinicians, who retain full authority to override algorithmic ordering based on clinical judgment, direct communication with donors, or institutional protocols. The system generates no autonomous dispatch actions. Patient identification on public-facing emergency tracking URLs is protected by cryptographic UUID tokens derived from request identifiers, preventing casual de-anonymisation of patient health emergencies from URL inspection. The immutable inventory audit ledger ensures that every blood unit reservation, release, and dispatch event is traceable to a specific user account, timestamp, and justification record—providing full accountability for regulatory and medicolegal review.

X. CONCLUSION AND FUTURE WORK

This paper presented LifeLink AI, a trustworthy hybrid clinical decision-support architecture for real-time emergency blood matching within the paradigm of Digital and Precise Medicine. The system addresses a well-documented and high-mortality failure mode in emergency healthcare logistics: the absence of automated, real-time, immunohematologically safe coordination between voluntary donors, licensed blood banks, and hospital emergency coordinators under acute temporal constraints.

The central and novel contribution of this work is the formally defined **two-layer safety contract**: an architecturally enforced boundary that confines deterministic immunohematological and statutory eligibility logic to Layer 1, and calibrated AI donor response propensity scoring to Layer 2, with no data flow path by which a Layer-2 AI output can influence a Layer-1 safety decision. This guarantee is architectural rather than probabilistic, making it independent of model performance degradation, distribution shift, or adversarial inputs. The pattern establishes a blueprint for safe AI integration in other high-stakes medical logistics domains where probabilistic models must coexist with non-negotiable deterministic safety invariants.

The calibrated logistic regression propensity model achieves ROC-AUC = 0.7521, Recall = 0.7528, and Brier Score = 0.2055 on the UCI BTSC benchmark under 5-fold stratified CV, confirming genuine and well-calibrated discriminative signal in the three-dimensional RFT feature space. The containerised multi-service platform passes 56 automated integration tests at 100% pass rate, with matching pipeline latencies of 47.2 ms (AI path, 95th pctl: 83.1 ms) and 12.4 ms (deterministic fallback, 95th pctl: 21.8 ms)—both well within the sub-second emergency dispatch budget.

Future Research Priorities: The most significant near-term directions are: (i) *Domain-adapted continual learning*—online retraining of the propensity model on live

LifeLink AI donor response telemetry using privacy-preserving federated optimisation across multiple deployment regions; (ii) *Learning-to-Rank formulation*—replacing heuristic weight vectors with LambdaMART or neural ranking models trained on historical emergency outcome sequences (whether contacted donors actually responded and donated), optimising normalised discounted cumulative gain (NDCG) on the ranked dispatch list; (iii) *FHIR R4 interoperability*—integrating with hospital EHR systems via HL7 FHIR R4 APIs to ingest blood type data from patient records automatically and cross-reference facility licenses against CDSCO/SBTC registries in real time; (iv) *Push notification dispatch*—integrating Firebase Cloud Messaging for automated, geofenced donor alerts with real-time response tracking; and (v) *Prospective clinical validation*—IRB-approved implementation study in regional trauma centres measuring primary endpoint of time-to-transfusion reduction against standard telephonic coordination baselines.

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